Kabilan Mahathevan

Computer Science PhD Student

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Profile

I am an alchemist! I believe in the law of equivalent exchange: I sacrifice complex compiler problems to gain ultimate runtime speed. My current mission is transmuting sparse tensor computation into something efficient and usable on heterogeneous hardware, abstracting away the chaos so users don't have to see the Truth behind the gate.

Areas of Interest & Expertise

Sparse Tensor Computation - High Performance Computing - Programming Languages & Compilers - Program Analysis - Machine Learning

Education

PhD Computer Science *(Virginia Tech)* **Blacksburg VA, USA** *2025-present*

- Advised by Prof. Kirshanthan Sundararajah.
- Working on sparse tensor compiler construction and validation.

BSc Computer Science and Engineering *(University of Moratuwa)* **Colombo, Sri Lanka** *2019-2024*

- CGPA: **3.92/4.2** (First Class)
- Dean's List Semesters: 7/8
- Batch Representative for 5th and 6th Semesters.

Publications

- **Splyce: SIMD Vectorization of Sparse Coiteration**
  Kabilan Mahathevan, Poorna Gunathilaka, and Kirshanthan Sundararajah
  PACT 2026 · *Proceedings of the 35th International Conference on Parallel Architectures and Compilation Techniques*
  Winner - Chameleon Reproducibility Challenge
- **Dialect-Agnostic SQL Parsing via LLM-Based Segmentation**
  Junwen An, **Kabilan Mahathevan**, and Manuel Rigger
  SIGMOD 2026 · *Proceedings of the ACM on Management of Data*, Vol. 4, No. 3, pp. 1–28
- **TENSURE: Fuzzing Sparse Tensor Compilers (Registered Report)**
  Kabilan Mahathevan, Yining Zhang, Muhammad Ali Gulzar, and Kirshanthan Sundararajah
  FUZZING 2026 · *Proceedings of the 5th International Fuzzing Workshop* · Co-located with NDSS 2026 · doi:10.14722/fuzzing.2026.23006
- **BugsInKube: A Collection of Reconciliation Bugs**
  Kabilan Mahathevan, Sivakajan Sivaparan, Tharsha Sivapalarajah, Sunimal Rathnayake, and Ridwan Shariffdeen
  APSEC 2024 · *Proceedings of the 31st Asia-Pacific Software Engineering Conference*, pp. 447–451 · doi:10.1109/APSEC65559.2024.00057

Research Experience

PhD Student *(Virginia Tech)* **Blacksburg, USA** *09/2025 - Present*

- Optimizing sparse tensor computation to guarantee correctness and efficiency across heterogeneous hardware systems.

Visiting Research Scholar *(TEST Lab, NUS)* **Singapore** *09/2024 - 05/2025*

- Worked with Prof. Manuel Rigger in increasing the performance of the mutation testing for SQL queries and dialect-agnostic AST for SQL queries using LLM-segmentation.

Fuzzing Cloud PaaS Systems *(University of Moratuwa)* **Colombo, Sri Lanka** *07/2023 - 06/2024*

- Developed a specialized fuzzing technique to systematically uncover reconciliation bugs within Kubernetes, as a vital component of my undergraduate final year project. The empirical data on the reconciliation bug in Kubernetes has been accepted to APSEC 2024.
- Advisors: Dr. Sunimal Rathnayake & Dr. Ridwan Shariffdeen

Teaching Experience

Teaching Assistant *(Virginia Tech)* **Blacksburg, USA**

- **CS4104 - Fall 2026 - Data and Algorithm Analysis**.
- **CS1064 - Fall 2025 - Introduction to Programming in Python**.

Teaching Assistant *(University of Moratuwa)* **Colombo, Sri Lanka** *07/2023 - 06/2024*

- CS3063 - Theory of Computing, CS2043 - Operating Systems, CS3043 - Database System

Professional Experience

Software Engineer *(Sysco LABS)* **Colombo, Sri Lanka** *06/2024 - 09/2024*

- Worked on a competition organizing and judging platform from backend to frontend.
- Implemented the unit test suite, regression testing, UI functional testing suite, and the relevant CI pipeline.

Software Development Engineer Intern *(The Software Practice)* **Cecil Court, Singapore** *01/2023 - 06/2023*

- Worked on **LEAP** project for Building And Construction Authority (**BCA**), Singapore.
- Worked with the front-end team to create a better and user-friendly interface.
- Created the entire Regression Testing Suite for the project under **Tony Tieu Ha Phong**.

Services

Artifact Reviewer

- 23rd USENIX Conference on File and Storage Technologies [**FAST'25**]
- Principles of Programming Languages [**POPL'27**]
- Principles and Practice of Parallel Programming [**PPoPP'27**]

Google Summer of Code

- Mentored SQLancer Project in GSoC 2025

Fellowships and Awards

- Student Travel Grant (PLDI-PLMW'26, VT GPSS'26, PACT'26)
- Pratt Fellowship (Virginia Tech, 2026/27)
- Mahapola Merit Scholarship (Sri Lanka, 2020-2024)

Skills

- Python
- C++
- LLVM/MLIR
- RDBMS
- Java
- Lean

Languages

- **English** [Fluent]
- **Tamil** [Native]